

No Dwarves for the Bear: An MR-Based Immersive Artwork Reinterpreting Modern Polish History

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Figure 1: Demonstration of the artwork implemented in a mixed reality (MR) environment.

ABSTRACT

"No Dwarves for the Bear" is a Mixed Reality (MR)-based immersive artwork that conveys the universal message of the value of freedom, inspired by motifs from modern Polish history, particularly the 1980s martial law period. This work is an artistic attempt, designed to allow the viewers to experience a virtual narrative superimposed on physical reality. The work adopts a childlike aesthetic, creating the impression of paper-made characters. Leveraging the Unity engine and the color passthrough feature of the Meta Quest 3s, the work establishes an interactive experience with virtual objects in a real-world environment. By utilizing these tools, this project is intended to go beyond simple message delivery, providing an artistic experience where the audience can personally discover and interpret the message within the work.

Index Terms: Virtual Reality, Mixed Reality, Storytelling, Cultural Exchange, Immersive Artwork

1 INTRODUCTION

Bekele et al. analyzed that the utilization of new and combined media enhances the way cultural heritage is experienced. Specifically, they noted that Augmented Reality (AR), Virtual Reality (VR), and MR technologies increase digital accessibility by enabling user-centered presentations [1]. In this regard, this work utilizes MR technology to directly project the narrative of oppression and struggle under the 1980s communist regime of modern Poland into the viewer's physical environment.

Furthermore, the project adopts an approach that allows viewers to interact freely with the artwork. By moving away from unidirectional information delivery and enabling viewers to freely observe and interact with the virtual space, the work intends to encourage autonomous interpretation of the underlying message.

The goal of this project is to refine the complex context of modern Polish history and sublimates its core theme, which is the universal value of freedom, into a highly immersive media art piece. Through this, the work seeks to demonstrate an expansion from a specific national historical narrative (Poland) into a broader artistic exploration of *freedom* as a fundamental human value.

2 IMPLEMENTATION AND INTERACTION FRAMEWORK

The artwork was developed using Unity 2022.3 LTS, Meta OpenXR SDK, and Meta Quest 3s Color Passthrough. The experiential workflow is as follows:

- The viewer wears the MR device (Meta Quest 3s), overlaying the virtual artwork onto the physical environment.
- The viewer freely explores the spatial integration of virtual assets and physical space.
- Aiming at an object with the controller, the viewer selects and picks up items of interest.
- A narration UI is triggered, revealing the historical narrative embedded within the manipulated object.

2.1 Interaction Mechanisms

The interaction mechanisms were designed to provide viewers with opportunities to actively experience and interpret the work:

- Anchor-based Manipulation System:** Picked-up objects are immediately parented to the controller's transform anchor, allowing them to be handled as a single unit. This enables viewers to easily perform various interactions, such as rotating the object from a distance

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using the joystick or bringing it closer for detailed observation, without spatial constraints.

- b. **Tag-based Narration System:** When an object is picked up, the system identifies the tag metadata stored in the object and displays related narration text as a UI. This encourages viewers to understand the hidden messages and narratives based on their active movements. Additionally, a billboard technique is applied by tracking the viewer's gaze to improve UI readability.
- c. **Gravity Control for Detailed Observation:** To allow viewers to conveniently observe the work from multiple angles, the Rigidbody's gravity option is disabled the moment an object is picked up, keeping it suspended in mid-air.

3 ARTISTIC FEATURES

3.1 Visual Metaphor

Upon wearing the MR device, viewers encounter a world composed of coarse models reminiscent of paper dolls. The presence of a giant teddy bear surrounded by small dwarves evokes a whimsical aesthetic similar to children's handicrafts, intended to pique the viewer's curiosity. However, these figures symbolize the history of Polish citizens' struggle against government oppression and authority. While the monochromatic, low-saturation color scheme may initially appear as a stylistic choice, it is a deliberate staging to potentially represent the controlled and oppressive social atmosphere of the martial law period. Viewers are provided with the opportunity to progressively decode these symbolic elements as the experience unfolds.

Unlike VR, which isolates the user within a purely synthetic environment [1][3], this work utilizes MR to fuse the virtual narrative with the current viewer's physical environment (e.g., a desk or room). To suggest that historical events are not isolated incidents of the past but *living memories* that continue to resonate in modern life, this spatial integration was utilized as a visual metaphor.

3.2 Message Delivery through Narrative Inversion

After entering the space, the viewer meets the paper-looking characters that, in a way, resemble toys that the viewer can interact with. Then, one by one viewer can pick up a dwarf or the bear, which is followed by a UI (Narration System) that appears in space, providing the historical narrative inherent to that object. Through this process, the viewer realizes that the "bear" symbolizes the violent military regime during the Polish martial law of the 1980s, while the "dwarves" represent civils that participated in resistance movements. As such, this narrative inversion, utilized alongside visual metaphors, was intended to leave a strong impression of the work's message on the viewer.

3.3 Storytelling through Proactive Participation

Viewers proactively interact with and observe the objects appearing in the work, gradually discovering the connections between the symbols that construct the narrative. For instance, the ambient sound, initially perceived as simple noise, is revealed through continued exploration to be a historical archive of the actual chants of citizens during the 1980s protests.

Building on the language of the cinematic cut, Barba suggests that AR and MR borrow certain formal elements from film but move away from the voyeuristic experience of film to embodied participation [2]. In this regard, this experiential approach, designed to encourage the audience to actively explore the work

and interpret its messages, was intended to convey that freedom is not given, but is a value earned through proactive participation.

4 CONCLUSION

This work demonstrates an attempt to refine the complex socio-political context of a specific national history of modern Poland and sublimate the universal human theme of the value of freedom into a highly immersive media art piece. Future research will need to evaluate the actual impact of this design strategy on emotional resonance and historical message delivery among real users. Ultimately, it is expected that this attempt will hold value as a tool for fostering an understanding of democracy and liberty beyond the boundaries of a specific nation.

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